

Bishwajit Sharma

bishwajitsharma444@gmail.com | (+91) 70051-68035

github.com/bishwajit-sharma101

linkedin.com/in/bishwajitsharma-in

PROFESSIONAL SUMMARY

Full-Stack Developer skilled in React.js, Node.js, MongoDB, and Socket.IO with experience building real-time and AI-powered web applications. Strong understanding of scalable backend systems, responsive UI/UX, and real-time communication architectures.

TECHNICAL SKILLS

Frontend & Languages: JavaScript (ES6+), TypeScript, React.js, Next.js, Tailwind CSS, Redux

Backend, Systems & AI: Node.js, Express.js, Socket.IO, MongoDB, Gemini API, Local LLMs (Gemma 4), Linux, Git

ENGINEERING PROJECTS

AstrixChat | Real-Time Communication Platform

React.js, Node.js, Express.js, MongoDB, Socket.IO

- Engineered a scalable full-stack messaging architecture using Node.js and Socket.IO, enabling low-latency, bidirectional communication and live user presence tracking.
- Benchmarked and optimized API performance using Autocannon to sustain 2,250+ requests per second (RPS) for message storage, maintaining <25ms average latency under high-concurrency load tests.
- Integrated a context-aware translation and grammar correction module optimized for ~1-second latency to maintain conversational realism and bridge cross-cultural communication gaps.

Kaevrix | Full-Stack Educational Esports Platform

React.js, Node.js, Socket.IO, Gemini API, Web Audio API

- Architected a 1v1 multiplayer educational platform using a Node.js/Socket.IO state machine to handle concurrent matchmaking, live spectator updates, and synchronized video playback with <2s drift tolerance.
- Integrated the Google Gemini API to dynamically generate context-aware quizzes, strategically masking API latency within a 5-second pre-match countdown.
- Developed a zero-asset client-side audio engine via the native Web Audio API, reducing bundle size and ensuring zero-latency sound effects for competitive mechanics.

Reina | Interactive 3D AI Companion Interface *React.js, Three.js, Gemma 4, VoiceVox, Mixamo, VRM*

- Developed an active 3D digital companion integrating local Large Language Models (Gemma 4) for autonomous, private, and personalized conversational logic.
- Utilized VRM models and Mixamo to map AI responses to dynamic 3D character movements and expressions directly within the browser.
- Engineered an expressive voice synthesis pipeline utilizing VoiceVox and Qwen3 TTS, optimized to run efficiently within local VRAM limits on consumer-grade hardware.

Education

Bachelor of Computer Applications (BCA)

Aryavart International University

Expected Graduation: 2026

- Relevant Coursework:** Operating Systems, Computer Networks, Data Structures & Algorithms, Database Management Systems (DBMS), Linux Administration.
- Performance:** CGPA 8.01/10 (Current)